



POLYGLOT WORKSHOP

INTRODUCTION TO LLMS

Tutors

Nicholas Kluge & Aniket Sen



WHO ARE WE / TEAM



THE TEAM / RESEARCH

Polyglot is an initiative to close the linguistic divide in NLP by developing efficient and accessible foundation models for low-resource languages (currently funded by [TRA Sustainable Futures](#)).

Nicholas → Engineering/Philosophy

Aniket → Physics

Sophia → Agriculture

Shiza → Comp. Sci.





THE TEAM / RESEARCH

Nicholas → LLM full stack

Aniket → HPC problem solver

Sophia → Environmental Alter Ego

Shiza → Nicholas's backup / upgrade







THE TEAM / SUPPORT

Julia → Philosophy

Lennard → Philosophy





THE TEAM / SUPPORT

Julia & Lennard → The ones who **actually do** all the hard work!!





2024



TeenyTinyLlama: tiny language models for Brazilian Portuguese



Tucano: Advancing Neural Text Generation for Portuguese



ViTucano: A Portuguese Vision Assistant

2026

LilTii: A 0.6B Bengali Language Model that Outperforms Qwen



Raising Bars, Not Parameters: LilMoo Compact Language Model for Hindi



Tucano 2 Cool: Better Open Source LLMs for Portuguese





WHO ARE YOU / PARTICIPANTS



AGENDA

1. A Very Brief (and Selective) History of NLP and LLMs

- ✓ What Is Natural Language Processing (NLP)?
- ✓ A High-Level Taxonomy of Language Models
- ✓ From Bag-of-Words to Embeddings
- ✓ *Attention is All you Need*
- ✓ Transformers and the Era of Scaling
- ✓ Alignment and Assistants

2. The Language Gap

- ✓ The Current Status Quo
- ✓ What Are Low-Resource Languages?
- ✓ What Keeps the Gap Open?

3. Polyglot Workshop

- ✓ What Will You Learn?
- ✓ What Outcomes Do We Aim to Achieve?



A VERY BRIEF (AND SELECTIVE) HISTORY OF NLP AND LLMS

From Bags-of-Words to Bags of GPUs



WHAT IS NATURAL LANGUAGE PROCESSING (NLP), AND WHY SHOULD WE CARE?

- ✓ Teaching machines to work with human language.
- ✓ One of the field's original moonshots → *"If computers can handle language, they can handle almost everything else."*
- ✓ **Dartmouth Summer Research Project on Artificial Intelligence (1950s)** → Language as a test of machine intelligence → *"If it talks like us, it thinks like us"* (maybe)
- ✓ Also, because language is ***an interface*** to knowledge and people (law, science, culture—all bottlenecked by language).

Jurafsky, D., & Martin, J. H. (2000). [Speech and language processing: An introduction to natural language processing, computational linguistics, and speech recognition.](#)



WHAT IS NATURAL LANGUAGE PROCESSING (NLP), AND WHY SHOULD WE CARE?

Another reason to try to “solve” language is that it is (*at least we thought it was*) a pretty tough challenge:

- ✓ **Language is: Ambiguous, Contextual, Social** (*very unlike math*)
- ✓ **Language isn't a puzzle with one solution**
- ✓ **Language is more like a moving target shaped by culture and power**
- ✓ **But what would “solving” NLP look like?** → *Understanding meaning?* → *Producing fluent text?* → *Passing as human?*

💡 *Fluency ≠ understanding* 💡

🧠 *Human Intelligence ≠ autocomplete* 🧠



EARLY NLP BREAKTHROUGHS

- ✓ Early NLP Breakthroughs → You could get “okayish” NLP capabilities by implementing straightforward rules and heuristics (e.g., counting words/n-grams).
- ✓ Rule-based systems, Symbolic grammars, Hand-written linguistic knowledge → my (Nicholas) first ever NLP/Chat system was like this (`n-gram decomposition + dictionary search`), and it was a pretty hard baseline to beat!
- ✓ **ELIZA**, the chatbot therapist → pattern matching + clever tricks → *ELIZA didn't understand you, but you felt understood.*
- ✓ However, they were *brittle, expensive* (in the cognitive sense), and they *didn't scale*.

“As we increase X, things improve ...”

Weizenbaum, J. (1966). [ELIZA—a computer program for the study of natural language communication between man and machine](#). *Communications of the ACM*, 9(1), 36-45.

A HIGH-LEVEL TAXONOMY OF LANGUAGE MODELS



- ✓ A language model is a system that can use patterns in language to perform some task (e.g., predict what comes next in a sequence of words, classify the sentiment of a sentence).
- ✓ **What are these tasks?** *Classification, Question Answering, Translation, Summarization, Ranking, Similarity, Text Generation* → Hugging Face's [task taxonomy](#).
- ✓ Same Input but Different Goals → The task is (usually) **defined by the output**, not the input.
- ✓ ⚠ A Deliberately Oversimplified Distinction ⚠ → **Text Classification & Text Generation** (not “true” but useful)



A HIGH-LEVEL TAXONOMY OF LANGUAGE MODELS



Text Classification → Fixed Set of Labels → Output Space is Small

>> E.g., Spam / Not spam, Positive / Negative.

Input Text →  → Distribution Over Classes / Label

A HIGH-LEVEL TAXONOMY OF LANGUAGE MODELS



Text Generation → Open-ended Text → Output space is much larger

>> E.g., Translation, Summarization, Chatbots

Input Text →  → Distribution Over a Vocabulary / Output Text

💡 **Fun Fact:** *Almost any NLP task can be reframed as text generation!*



FROM BAG-OF-WORDS TO EMBEDDINGS

- ✓ **For specific tasks**, counting words isn't enough (*although we still use simple counting models for a lot of stuff*)
- ✓ Machines don't read — they compute → Language must become numbers → **How much structure** do those numbers capture?

🧠 Bad representations →  → smart models still fail 🗣️

“Dog bites man.” ≠ “Man bites dog.”



FROM BAG-OF-WORDS TO EMBEDDINGS

Text → Tokens → Numbers → Math!

“Man bites dog.” → f → [‘man: 42’, ‘bites: 69’, ‘dog: 1337’]

[42, 69, 1337] →  → [666, 42]

[666, 42] → f^{-1} → *“Bad Man”*



FROM BAG-OF-WORDS TO EMBEDDINGS

Tokenization is already a modeling choice.

- ✓ Tokens can be words ("cat"), subwords ("un", "believ", "able"), Characters ("c" "a" "t"), Bytes (0x63 0x61 0x74)
- ✓ We will talk more about tokenization tomorrow, since it is a **vital part of LLM development**, with many **caveats** when working with low-resource languages.

Too big → slow & sparse

Too small → loses meaning



FROM BAG-OF-WORDS TO EMBEDDINGS

The Bag-of-Words approach: represents text as a “bag” of tokens (*ignores grammar and order*).

- ✓ Each document → a vector where each dimension counts a specific token. Simple and fast, but **loses meaning** (“*Dog bites man*” = “*Man bites dog*”).
 - ✗ No Word Order
 - ✗ No Meaning (“good” and “excellent” are represented as unrelated dimensions)
 - ✗ Very High-dimensional & Sparse

Scaling Bag-of-Words ≠ Better Performance (no *true* scaling)



FROM BAG-OF-WORDS TO EMBEDDINGS

- ✓ But Bag-of-Words models are still useful, and we should not dunk on them too much.
- ✓ **Text classification:** Bad-word filter → Count swear words → flag text → Fast & Easy (easy to bypass)
- ✓ **Text generation:** Markov models for autocomplete → Predict next word from last n words → Fast & Easy (sounds OK... until it doesn't: *"I want to eat pizza pizza pizza pizza ..."*)

💡 *We need a way to encode more information into these units of representation/tokens.* 💡

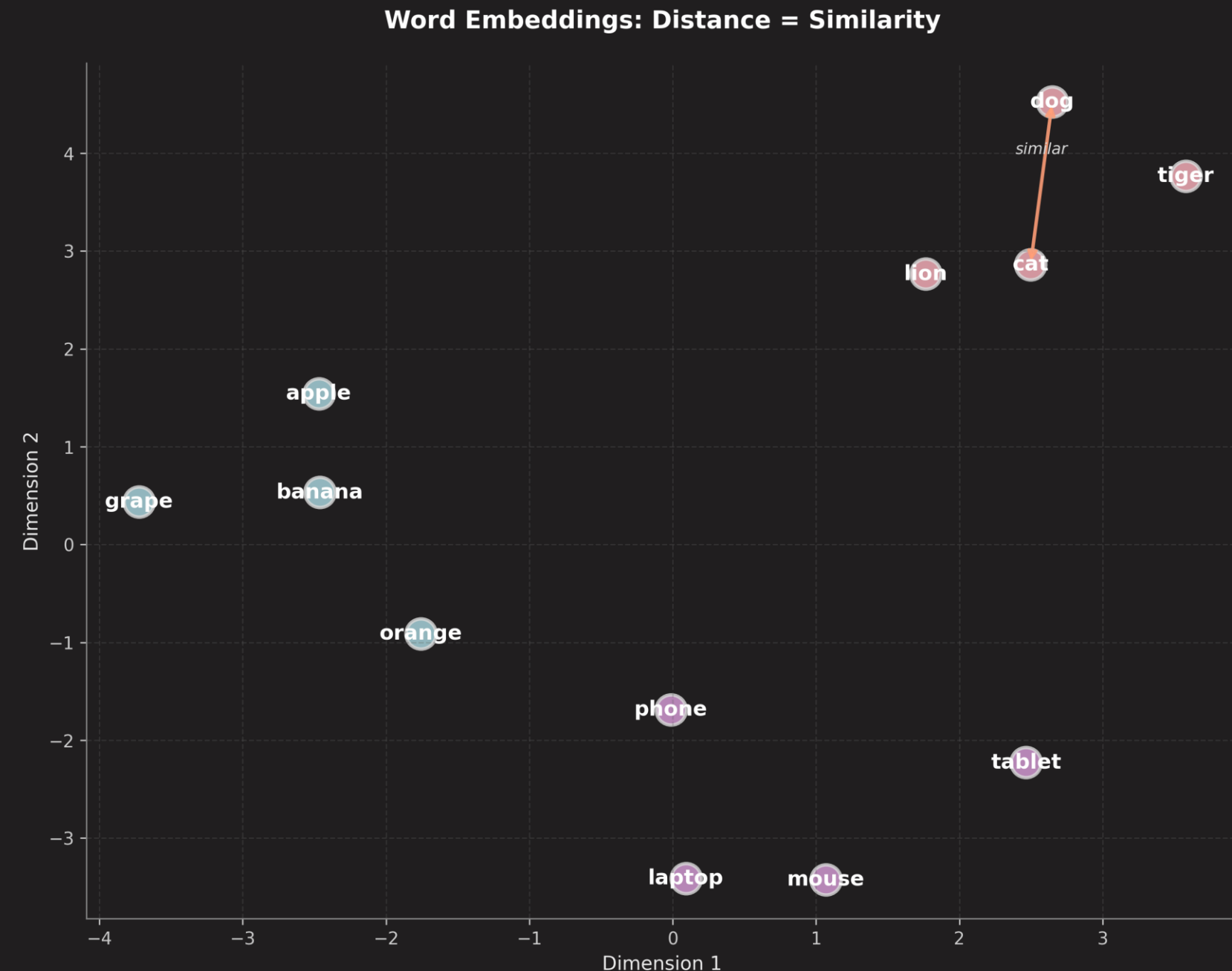
FROM BAG-OF-WORDS TO EMBEDDINGS



The key shifts that bring us to Embeddings:

- ✓ Tokens are points in space
- ✓ Meaning = geometry
- ✓ No more One-hot vectors
- ✓ Dense vectors with low dimensionality
- ✓ Feature vectors learned from data
- ✓ Distance now means similarity

Word2Vec: The Canonical Example



Mikolov, T., Chen, K., Corrado, G., & Dean, J. (2013). [Efficient estimation of word representations in vector space](https://arxiv.org/abs/1301.3781). arXiv preprint arXiv:1301.3781.

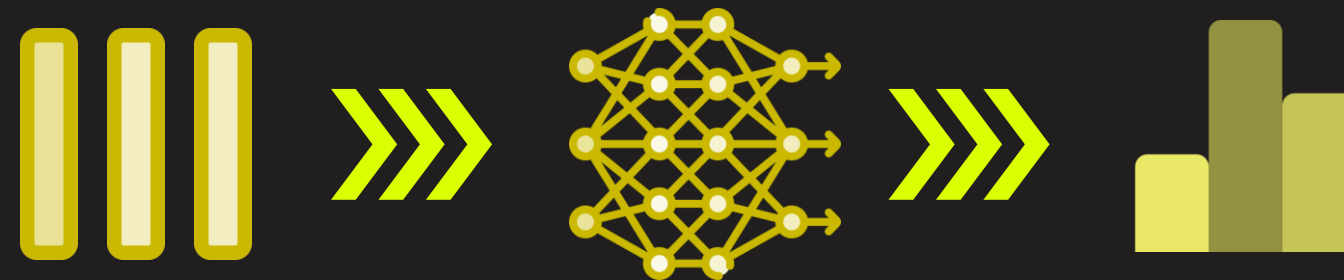


FROM BAG-OF-WORDS TO EMBEDDINGS

Embeddings (by themselves) unlock many things because **meaning becomes measurable**.

-  **Retrieval** (semantic search)
-  **Similarity** (nearest neighbors)
-  **Clustering** (topics, senses)

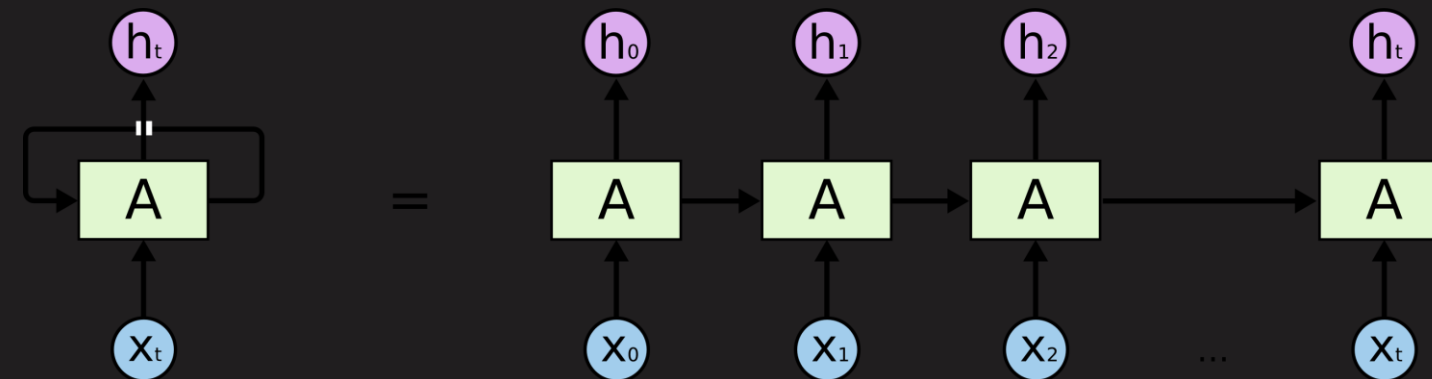
With these rich embeddings of features, we can now use them as the atoms of more elaborate forms of computation (aka, neural networks).





FROM BAG-OF-WORDS TO EMBEDDINGS

- ✓ **RNNs & LSTMs:** language is a time series → Order matters again.
- ✓ Process tokens one by one (**sequential modeling, not parallel**).
- ✓ Maintain a hidden state (**fixed-size**).
- ✓ Context is compressed memory (**unlimited context length, but compression is lossy**).



Schmidt, R. M. (2019). [Recurrent neural networks \(rnns\): A gentle introduction and overview](#). *arXiv preprint arXiv:1912.05911*.

Hochreiter, S., & Schmidhuber, J. (1997). [Long short-term memory](#). *Neural computation*, 9(8), 1735-1780.



FROM BAG-OF-WORDS TO EMBEDDINGS

RNNs & LSTMs were a significant step forward — **but not the last:**

- ✓ Historically, they were the **state-of-the-art pre-Transformer era**.
- ✓ **Google Neural Machine Translation ([GNMT](#))** was a system developed by Google (introduced in November 2016) that powered Google Translate. It was an encoder-decoder LSTM-based architecture with 8 1024-wide layers each and a simple 1-layer 1024-wide feedforward attention mechanism connecting them (still within the sub-billion-parameter range).

Core problems:

- ✗ Vanishing/Exploding gradients
- ✗ Slow training (no parallelism)
- ✗ Long-range dependencies are hard



“ATTENTION IS ALL YOU NEED”

(Yes, *that* paper.)

- ✓ Got rid of recurrence
- ✓ Popularized the **self-attention mechanism**
- ✓ Proposed the original **Transformer architecture**
- ✓ And later, we would find out that we were given something that (*actually*) scales

The best visualization ever created of a decoder-only transformer.

Attention Is All You Need

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Abstract

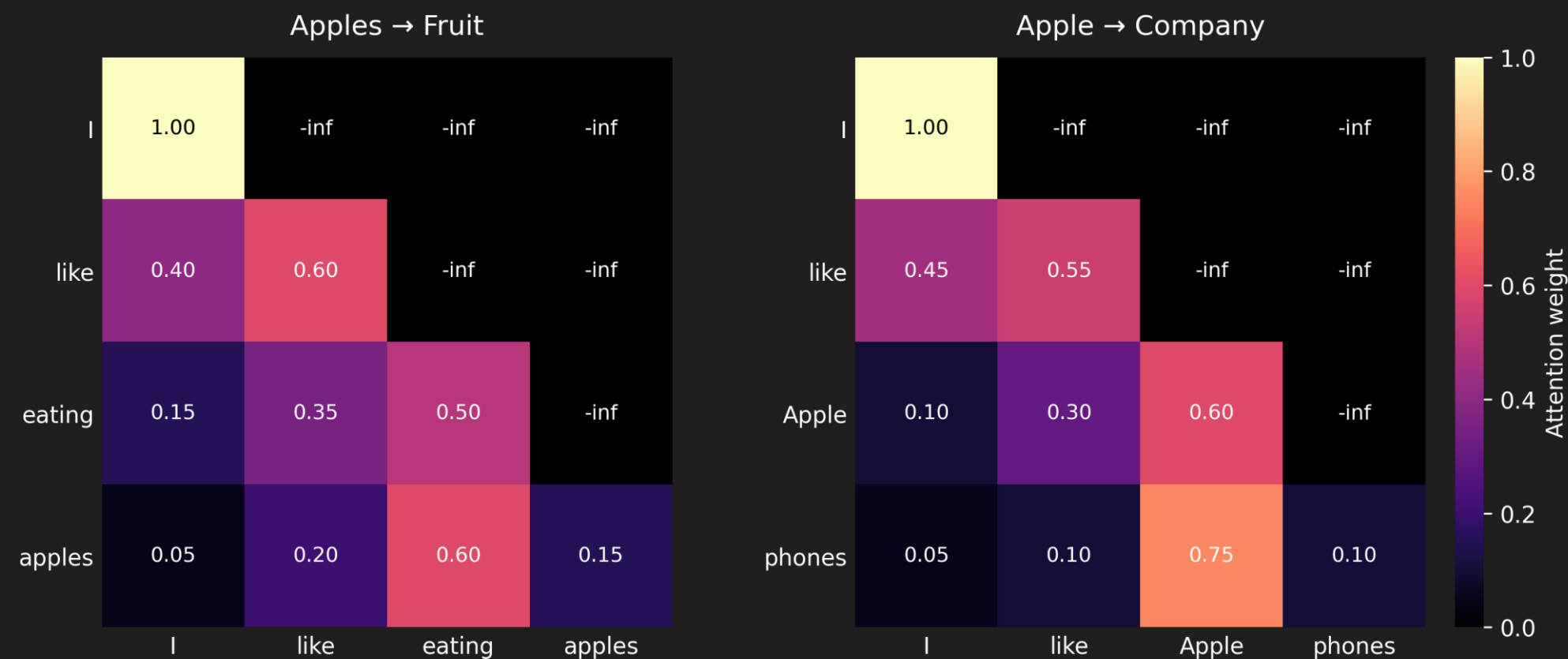
The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the existing best results, including ensembles, by over 2 BLEU. On the WMT 2014 English-to-French translation task, our model establishes a new single-model state-of-the-art BLEU score of 41.8 after training for 3.5 days on eight GPUs, a small fraction of the training costs of the best models from the literature. We show that the Transformer generalizes well to other tasks by applying it successfully to English constituency parsing both with large and limited training data.

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., ... & Polosukhin, I. (2017). [Attention is all you need](#). *Advances in Neural Information Processing Systems*, 30.



“ATTENTION IS ALL YOU NEED”

- ✓ What *is* Attention? → Attention is a way to allow for **feature vectors to “talk” to each other** and exchange information.
- ✓ Attention = **context-aware feature mixing**.





TRANSFORMERS AND THE ERA OF SCALING

- ✓ Before, models had hit a wall, but now performance improves predictably with more **data**, more **parameters**, more **compute**.
- ✓ This is *not* true for many other neural architectures.
- ✓ And now we have JARVIS.

 **Self-Attention**
(Mixes information)

 **MLPs**
(Applies non-linearity)

+ **Residual connections**
(Adds depth)

 **Normalization**
(Adds stability)



Kaplan, J., McCandlish, S., Henighan, T., Brown, T. B., Chess, B., Child, R., ... & Amodei, D. (2020). [Scaling laws for neural language models](https://arxiv.org/abs/2001.08361). *arXiv preprint arXiv:2001.08361*



ALIGNMENT AND ASSISTANTS

Currently, the modern assumption is: ***big enough models $\approx$ programmable behavior.***

- ✓ One model (**foundation**) → many **personas** 🧑 (e.g., tutor, coder, poet, analyst)
- ✓ Given the proper conditioning, we can make these foundations ***do almost anything.***

How?

- ✓ Training = Capability (World Knowledge)
- ✓ Alignment = Control (Application of said Knowledge)

💡 **Fun Fact:** Alignment research was one of the main forces responsible for giving us ChatGPT!

Bommasani, R. (2021). [On the opportunities and risks of foundation models](#). arXiv preprint arXiv:2108.07258.



ALIGNMENT AND ASSISTANTS

In other words, by learning how to “*teach machines what humans want*”, we learned how to create systems that can (*to a certain extent*) be extensions of our volition.

Systems that learn to conform to:

- ✓ Preferences (what we **like** ...)
- ✓ Norms (what we **ought** ...)
- ✓ Safety (what we **shouldn't** ...)

🧠 Even if we don't all agree on these things (*and maybe we never will*), it's **undeniable that we have come a long way from ELIZA to having a small oracle that we can access at any time for almost anything.**

Ouyang, L., Wu, J., Jiang, X., Almeida, D., Wainwright, C., Mishkin, P., ... & Lowe, R. (2022). [Training language models to follow instructions with human feedback](#). *Advances in neural information processing systems*, 35, 27730-27744.

... AND AGENTS

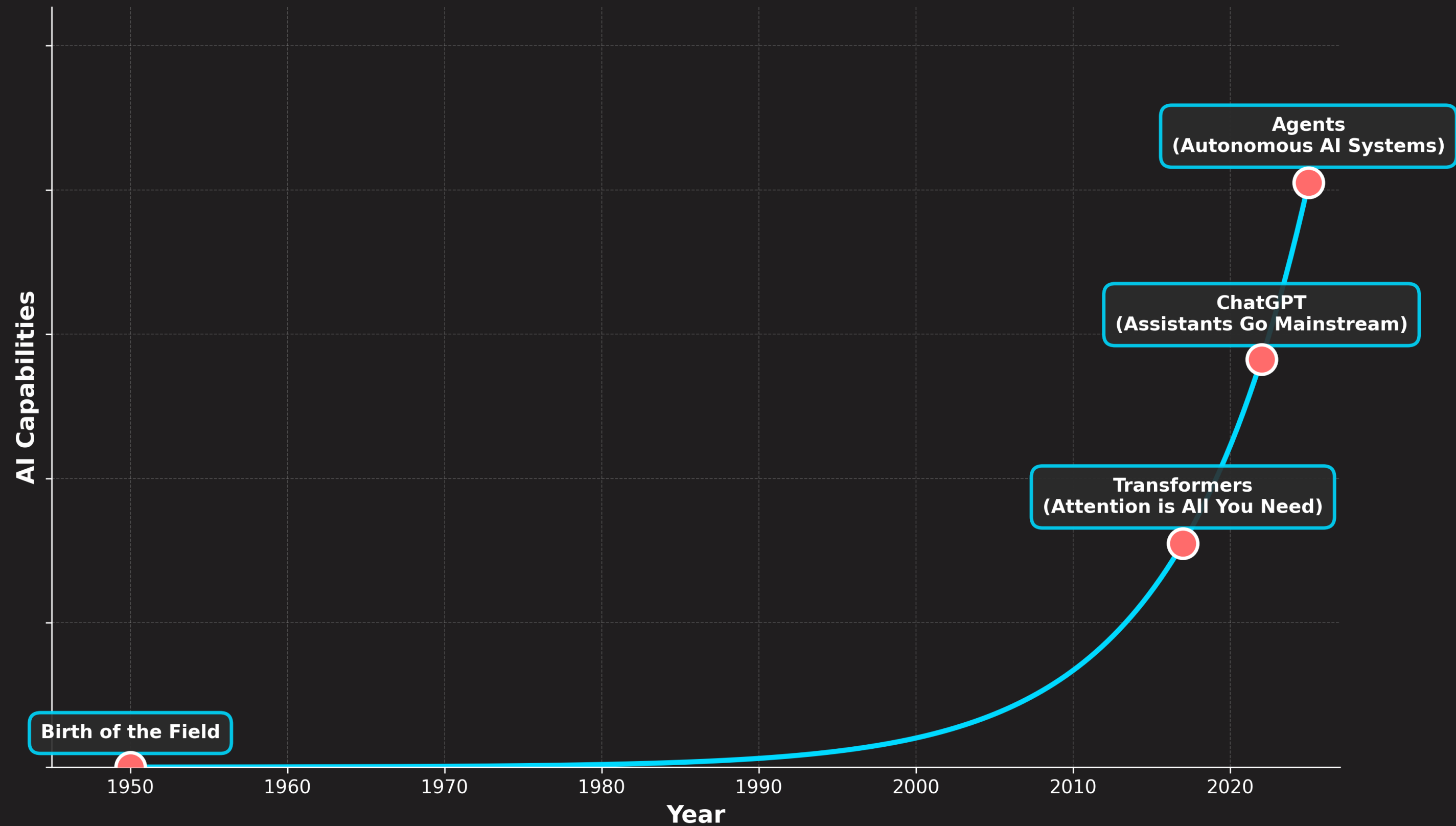


And now, we even have agents that can go out and **do tasks on their own**. Neural networks (aka, piles of matrices and learned parameters) that:

- ✓ Take Actions (create files and execute scripts)
- ✓ Use Tools (call APIs)
- ✓ Maintain a State (have long-term memory)
- ✓ Reason (self-steer themselves to be more capable)

Yao, S., Zhao, J., Yu, D., Du, N., Shafran, I., Narasimhan, K. R., & Cao, Y. (2022, October). [React: Synergizing reasoning and acting in language models](#). In *The eleventh international conference on learning representations*.

... AND AGENTS





BREAK



THE LANGUAGE GAP

The Long Tail of Human Language



THE CURRENT STATUS QUO

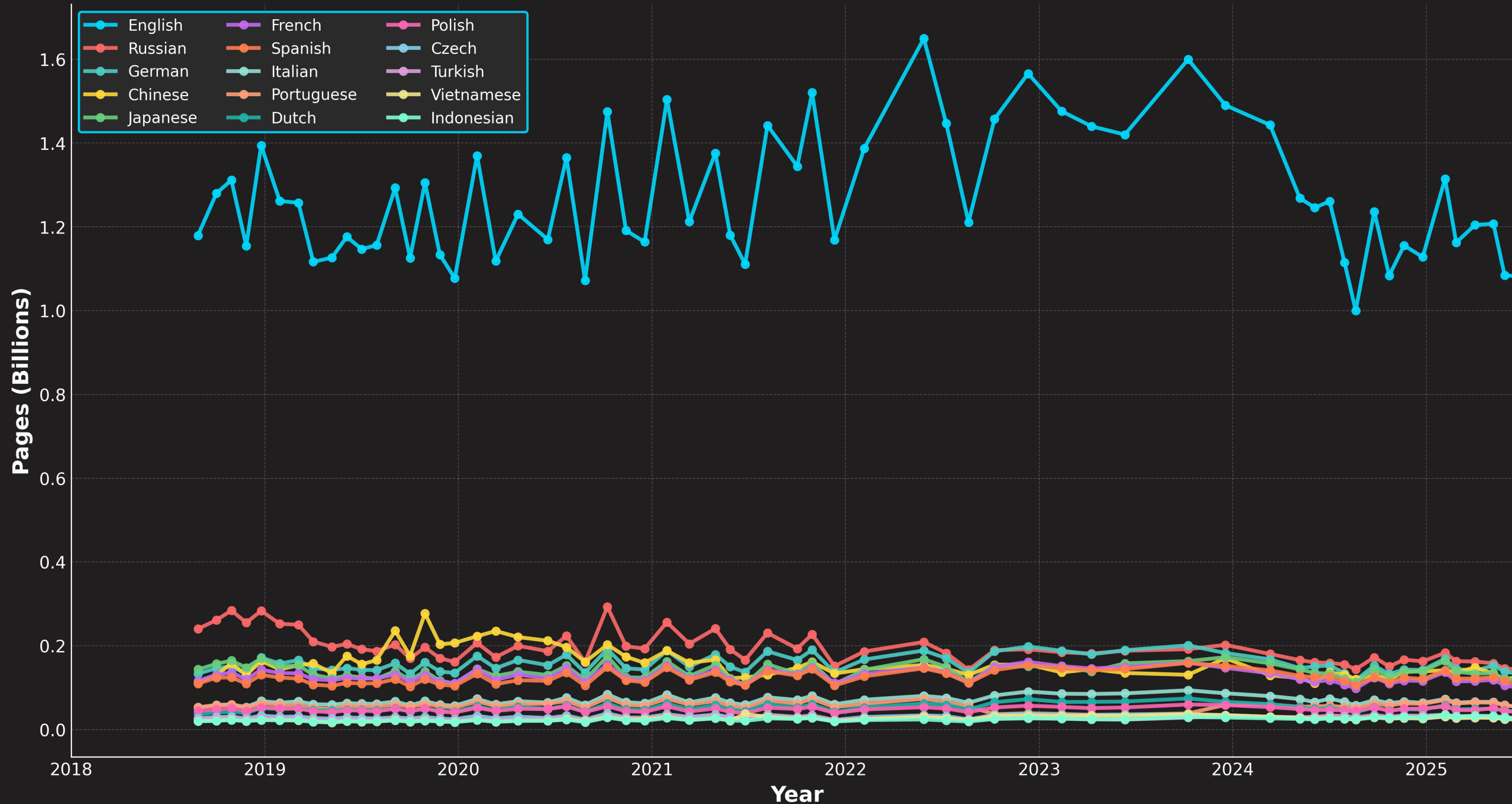
1. Current LLMs **struggle** when having to deal with **languages from the long tail**.
2. Most progress happens in **high-resource languages** (e.g., English).
3. Power is concentrated. A handful of labs define what **“state-of-the-art”** is.
4. For many languages and cultures, **time froze**. Many languages still **lack models that match the 2018–2019 systems**.
5. And if you add the restriction of **“open-source systems,”** then it is even worse ...

💡 A very cool resource for research: [Endangered Languages Project](#).

Pratik Joshi, Sebastin Santy, Amar Budhiraja, Kalika Bali, and Monojit Choudhury. 2020. [The State and Fate of Linguistic Diversity and Inclusion in the NLP World](#). In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 6282–6293, Online. Association for Computational Linguistics.



Evolution of Language Presence on the Web (2018-2025)





WHAT ARE LOW-RESOURCE LANGUAGES?

Source	What “resources” mean	How languages are labeled
Joshi et al. (2020)	<i>Presence of language resources (corpora, Wikipedia, labeled datasets)</i>	<i>6 qualitative classes (Class 0–5), no fixed size thresholds</i>
NLLB-200 (2024)	<i>Amount of parallel data (aligned sentence pairs)</i>	<i>Numeric cutoffs (e.g. <100k = very low-resource)</i>
Hedderich et al. (2021)	<i>Task-dependent (monolingual, labeled)</i>	<i>No fixed taxonomy — resource level is contextual</i>
Ranathunga et al. (2021)	<i>Whatever data is relevant for the task</i>	<i>Implicit, ad-hoc definitions (“a few thousand sentences”, etc.)</i>

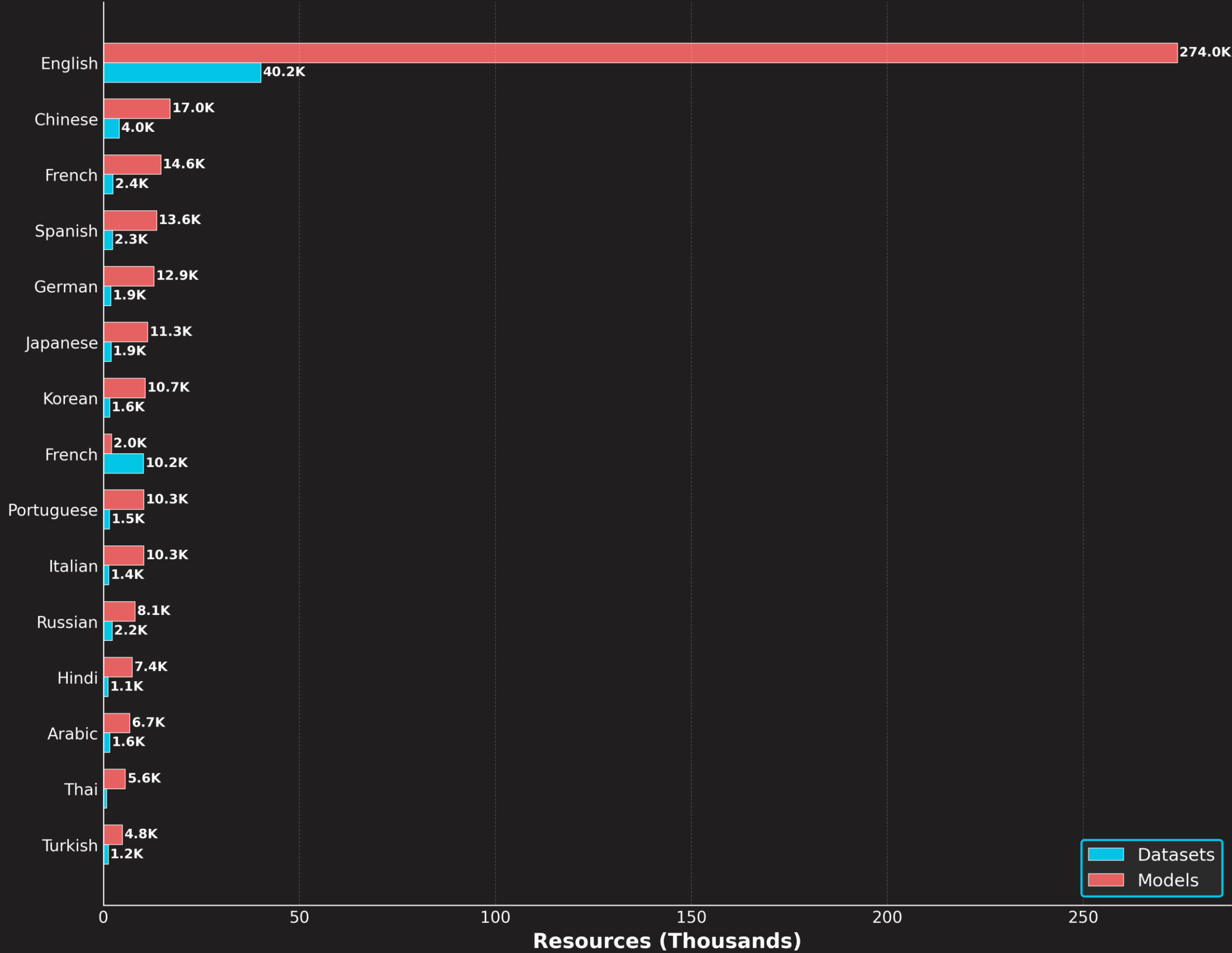


WHAT ARE LOW-RESOURCE LANGUAGES?

We can definitely define “low-resource” in the context of (obviously) resources:

- ✗ **Data scarcity** → *both analog and digital — analog adds extra friction*
- ✗ **Infrastructure** → *hardware, connectivity, stable energy grids*
- ✗ **Missing Evaluations & Baselines** → *hard to tell whether progress is real*
- ✗ **Limited know-how** → *expertise is concentrated; open research helps, but unevenly*
- ✗ **Capital as the hidden bottleneck** → *money can mitigate most of the above*

Distribution of Hugging Face Resources by Language
(Top 15 Languages)

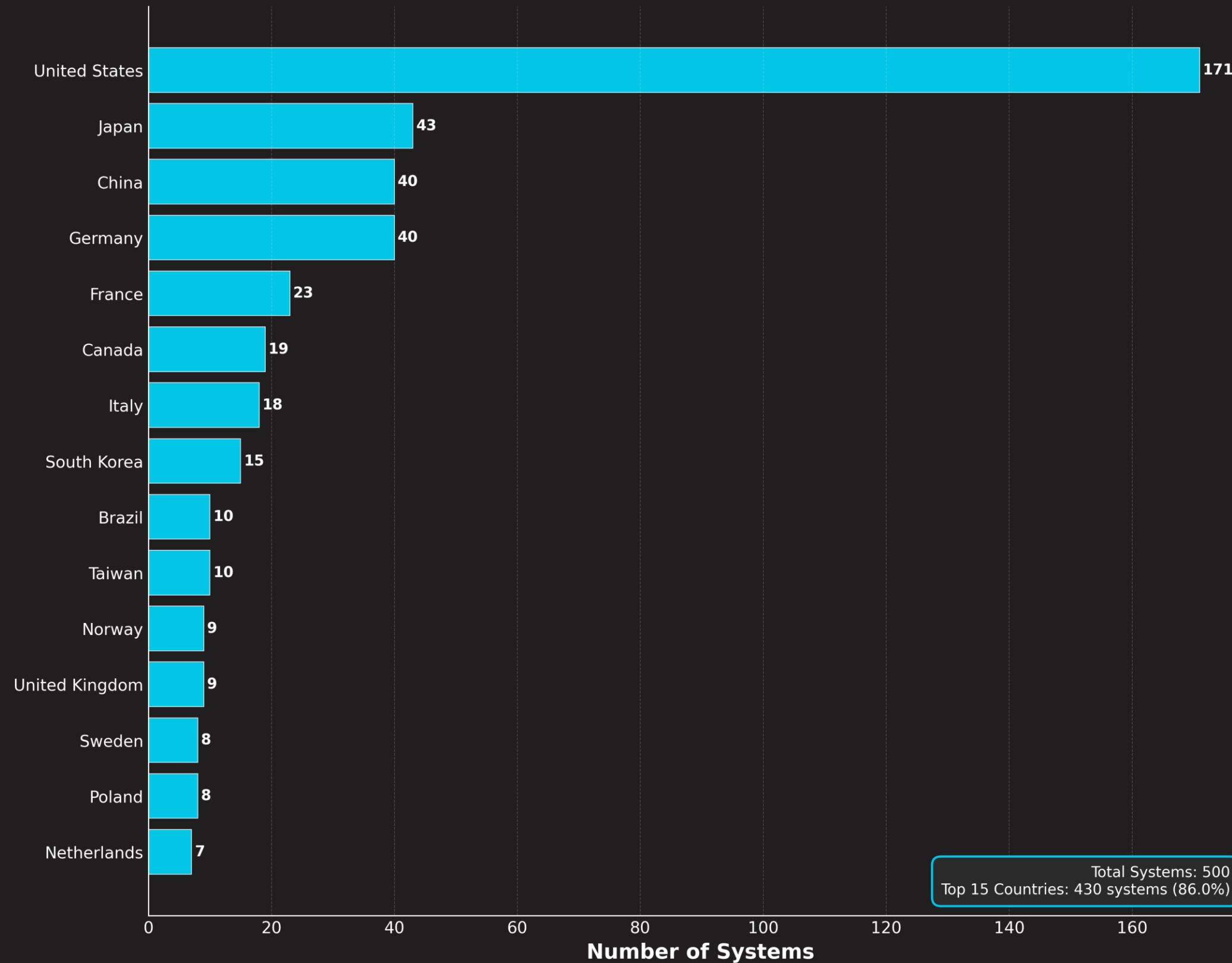


[Source → Hugging Face](#)

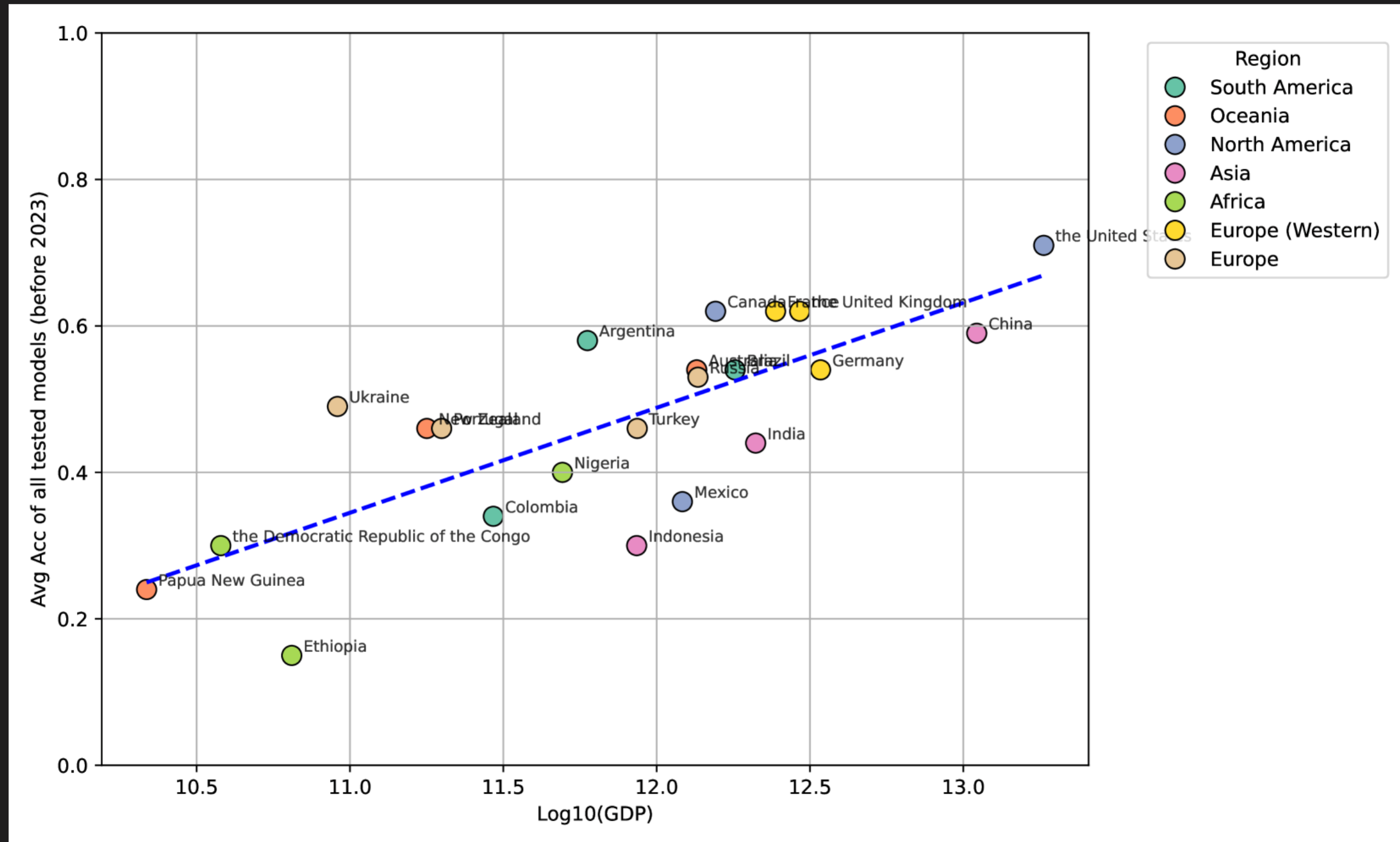
POLYGLOT



TOP500 Supercomputers by Country (November 2025) Top 15 Countries



[Source → TOP500 Stats \(Nov 2025\)](#)



Almeida, T. S., Bonás, G. K., Santos, J. G. A., Abonizio, H., & Nogueira, R. (2025). [TiEBE: Tracking Language Model Recall of Notable Worldwide Events Through Time](#). arXiv preprint arXiv:2501.07482.



WHAT KEEPS THE GAP OPEN?

I believe there are three bottlenecks, ***all structural***.

- 💡 **English dominates the web:** *No Data → No Model → No Representation*
- 💡 **Infrastructure is expensive:** *access to compute = access to “progress”*
- 💡 **Know-How is rare:** *Irreproducible, Under-documented, branded as “transparent” or “sovereign”*

We (the Polyglot team) have no secret data to share and no infrastructure to give.

But we *do* have know-how — and that’s what this workshop is about. Our fight is against the **know-how bottleneck**.



DO WE NEED A BREAK



POLYGLOT WORKSHOP

Open-Source against the know-how bottleneck



WHAT THIS WORKSHOP ISN'T ABOUT

✗ Not about:

>> **Train your own GPT in 1 day** → *LLM development (science) takes time!!!*

✓ Is about: *Imagine you had the data and the compute ...*

>> *What should you do first?*

>> *What shouldn't you do?*

>> *Where do most projects fail — and why?*

>> *From recipes → Reasoning*



WHAT THIS WORKSHOP ISN'T ABOUT

Not about **exploring/re-designing every detail of the stack**, because the stack is **TOO deep**.



We will necessarily **move fast**. If the university ever turns this into a **6-month course**, we'll happily **slow down**.



WHAT THIS WORKSHOP ISN'T ABOUT

So, we assume you already know (and hence, we will not be reviewing) things like:

- ✓ The core basics → Python [[1](#)], Git [[1](#), [2](#)], SSH [[1](#), [2](#)], Linux Basics [[1](#), [2](#)]
- ✓ You are already familiar with neural networks [[1](#), [2](#), [3](#)] and PyTorch [[1](#)]
- ✓ You know what a Transformer is, and hopefully have implemented it once [[1](#), [2](#), [3](#)]
- ✓ You have some experience in running inference on LLMs, and perhaps have even trained some [[1](#), [2](#)]

💡 ALL hyperlinks will take you to free resources to help you learn/refresh a given subject.



WHAT WILL YOU LEARN

Without HPC (High-Performance Computing), there are no trillion-token datasets, no *multi-GPU training*, no *scaling*. LLMs are as much a systems problem as an ML one.

Introduction to HPC → Jan Steiner (Uni Bonn / HPC@HRZ)

- ✓ The ABCs, i.e., *SLURM*, *Jobs*, *Failing Gracefully*, *DL choices that make GPUs go brrrrrrrrrr*
- ✓ How does an HPC cluster operate? How are resources shared?
- ✓ How does job submission work?
- ✓ What are some best practices when submitting jobs? How to have “good” fault tolerance?
- ✓ From a Deep Learning perspective, what are the things you should know about infrastructure?



WHAT WILL YOU LEARN?

Working with large-scale datasets and learning to convert web junk into high-quality data (for whatever language you are interested in) is most of the battle when developing an LLM.

Working with Text Datasets → Shiza Fatima & Nicholas Kluge

- ✓ The ABCs, i.e., *downloading, extracting, formatting, filtering, deduplication, and synthetic creation.*
- ✓ Working with Common Crawl dumps and HF datasets.
- ✓ Data filtration at scale (i.e., LID, quality checks, learned filters)
- ✓ Synthetic Data Generation
- ✓ Tokenization (nightmares and horrors). How to train and evaluate them?



WHAT WILL YOU LEARN?

Training LLMs is the marathon of ML. Success depends less on “*perfect data*” and more on knowing what you’re doing—and adapting when things inevitably go wrong.

Pretraining 101 → Nicholas Kluge & Aniket Sen

- ✓ The ABCs, i.e., *architectures, frameworks, parallelism, ablations, and evaluations*.
- ✓ Good architectural design practices. What are people using?
- ✓ Megatron, Nanotron, TorchTitan, Accelerate ... What to use?
- ✓ DDP, FSDP, FSDP2, HSDP, TP, PP, CP ... What to use?
- ✓ Ablations! Perhaps the most valuable thing to know.
- ✓ Evaluations. What makes a good eval? How to *evaluate* evals, and how to create them.



WHAT WILL YOU LEARN?

Pretraining $\neq$ usefulness. Post-training and alignment turn raw capability into practical behavior. This is where instruction-following, preference learning, reasoning, and tool use are taught and refined.

Post-Training and Alignment → Florian Mai & Nicholas Kluge

- ✓ The ABCs, i.e., *data types, templating, SFT, DPO, and more*
- ✓ Data/dataset types for conditional text generation.
- ✓ Templating (back to nightmares and horrors)
- ✓ Supervised Fine-Tuning (Instruct and Beyond) and Preference Learning (DPO and its variants)
- ✓ Evaluations



WHAT WILL YOU LEARN?

All of this can be extremely wasteful if you are not tracking what you are doing. Measuring, reporting, and optimizing your experiments is what we, as practitioners of the field, can do to promote sustainability.

Environmental Practices to LLM Development → Sophia Falk

- ✓ The ABCs, i.e., *measuring efficiency, energy consumption (CO₂eq) tracking, beyond carbon*
- ✓ How to measure if you are using all that the hardware can give (we never are 😞)
- ✓ Tracking energy consumption to infer equivalent carbon emissions
- ✓ Beyond Carbon! What else powers our GPU-intensive practices?



WHAT WILL YOU LEARN?

TRA (Transdisciplinary Research Areas) is all about (you guessed) transdisciplinarity. On our final day, we will have a talk exploring how the type of knowledge we have can make us extremely useful in other fields.

LLMs for Humanities → Alexander Ermakov

- ✓ Applications for LLMs in the Humanities
- ✓ Egyptology!
- ✓ Archaeology!
- ✓ And 1000 different things you can do with an HPC+ LLMs!



WHAT WILL YOU LEARN

1. If all of this feels overwhelming, that's a **healthy and sane reaction**.
2. The modern LLM development stack is deep—mastering it as a single person is a feat few achieve. You won't become an expert in a one-week workshop.
3. But if all goes according to plan, you'll leave **ready to start the kind of research we do here**—and, hopefully, avoid some of the 1001 mistakes we made at the beginning.

💡 Some resources (books) you can download → [here!](#)



WHAT OUTCOMES DO WE AIM TO ACHIEVE?

By the end of the workshop:

- ✓ You should be able to reproduce this kind of work (**1000 of low-hanging fruit**)
- ✓ For **your own contexts**
- ✓ And **your own languages**

We **cannot** tell you what works better for your context/language, but we can try to give you a recipe to **find that out yourself**.



NEXT ON THE AGENDA



KEYNOTE SPEAKERS

13:00 — 16:00

Marcellus Amadeus (Portuguese)
SoberaIA

Rabindra Nath Nandi (Bengali)
Verbex AI



MARVIN TOUR

17:00 — 18:00

Jan Steiner

Uni Bonn / HPC@HRZ



LUNCH TIME